Role of dietary n-3 polyunsaturated fatty acids in type 2 diabetes: a review of epidemiological and clinical studies.

The worldwide increasing prevalence of type 2 diabetes mellitus (T2DM) poses an immense public health hazard leading to a variety of complications such as cardiovascular diseases, nephropathy and neuropathy. Diet, as a key component of a healthy human lifestyle, plays an important role in the prevention and management of T2DM and its complications. The dietary n-3 polyunsaturated fatty acids (PUFAs) have been associated with various favourable functions such as anti-inflammatory effects, improving endothelial function, controlling the blood pressure, and reducing hypertriglyceridemia and insulin insensitivity. According to some epidemiological studies, a lower prevalence of T2DM was found in populations consuming large amounts of seafood products, which are rich in n-3 PUFAs. However, the evidence on the relation between fish intake, dietary n-3 PUFAs, and risk of T2DM is controversial. Therefore, this paper aimed to review the epidemiological and clinical studies on the role of dietary n-3 PUFAs in T2DM. Also, the limitations of these studies and the need for potential further research on the subject are discussed.